WEEK-3

Basic Plots for Data Exploration (Use Iris dataset):

1. Generate box plot for each of the four predictors.

2. Generate box plot for a specific feature

3. Generate histogram for a specific feature

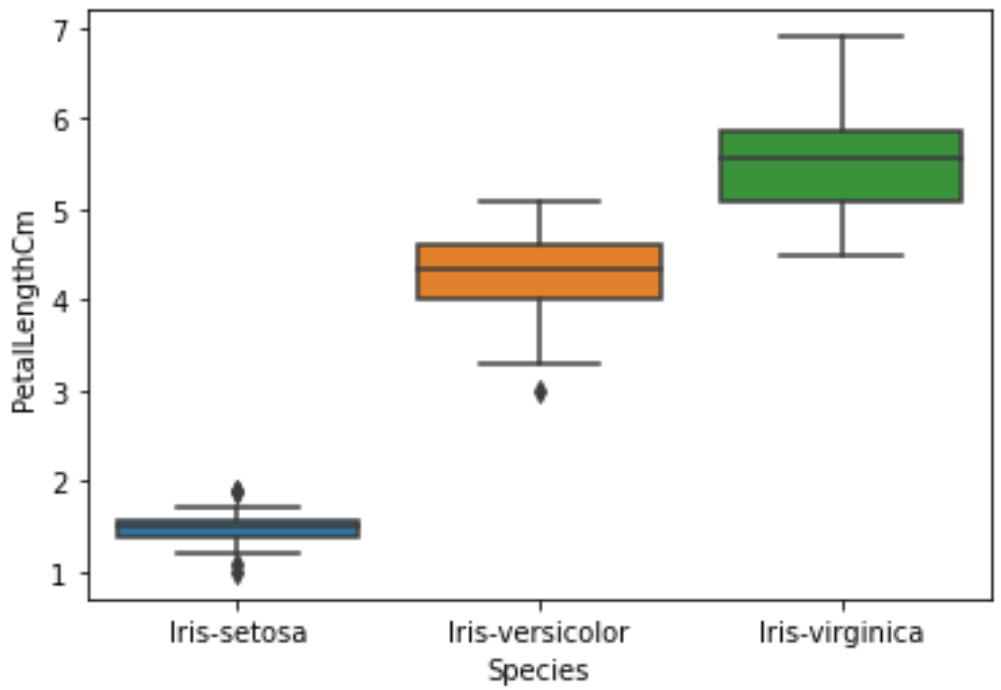
4. Generate Scatter plot of petal length vs. sepal length

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
#from statistics import median

#pip install matplotlib==3.2.2

df=pd.read_csv('iris.csv')
df.head()
```

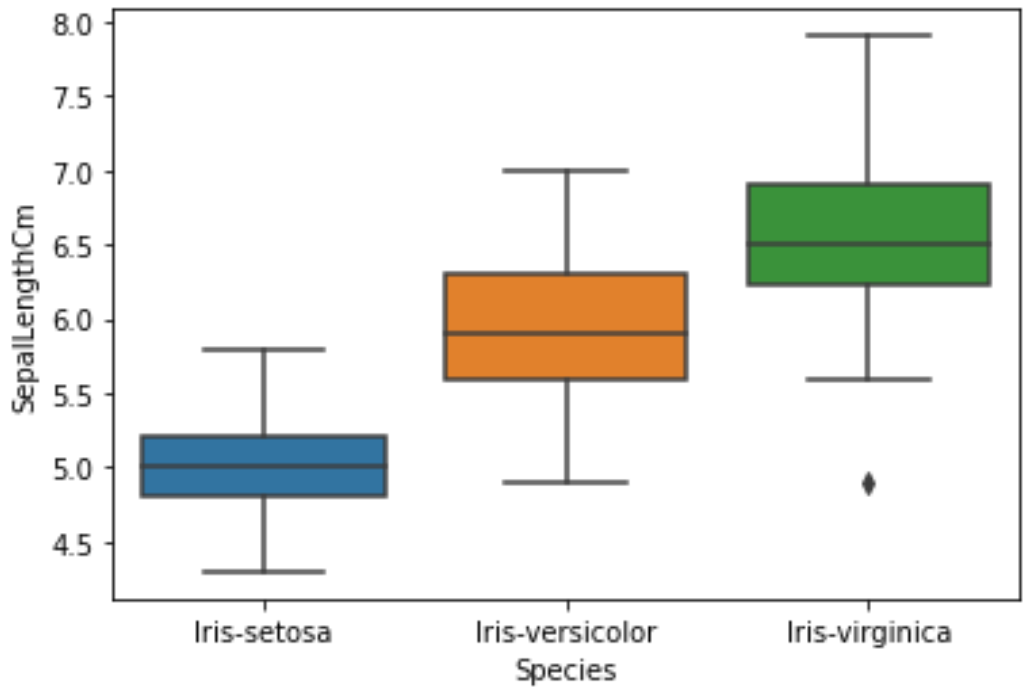
```
   Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm      Speci
es
0   1            5.1           3.5            1.4           0.2  Iris-seto
sa
1   2            4.9           3.0            1.4           0.2  Iris-seto
sa
2   3            4.7           3.2            1.3           0.2  Iris-seto
sa
3   4            4.6           3.1            1.5           0.2  Iris-seto
sa
4   5            5.0           3.6            1.4           0.2  Iris-seto
sa
```

Box plot for each of the four predictors.

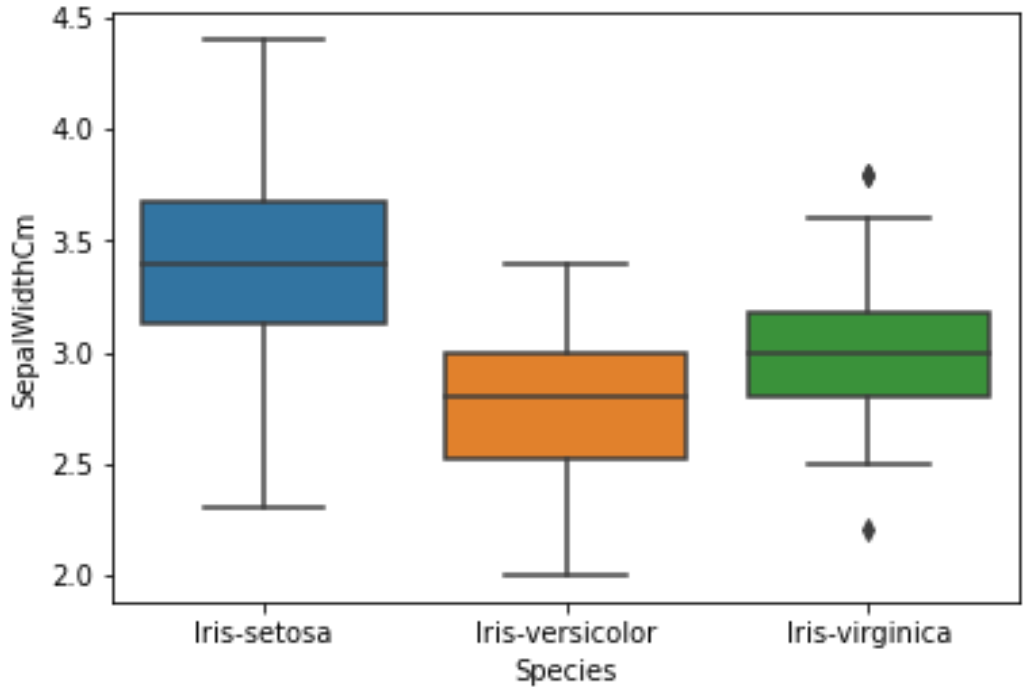
```
#Box plot for PetalLength
sns.boxplot(x='Species',y='PetalLengthCm',data=df)
plt.show()
```



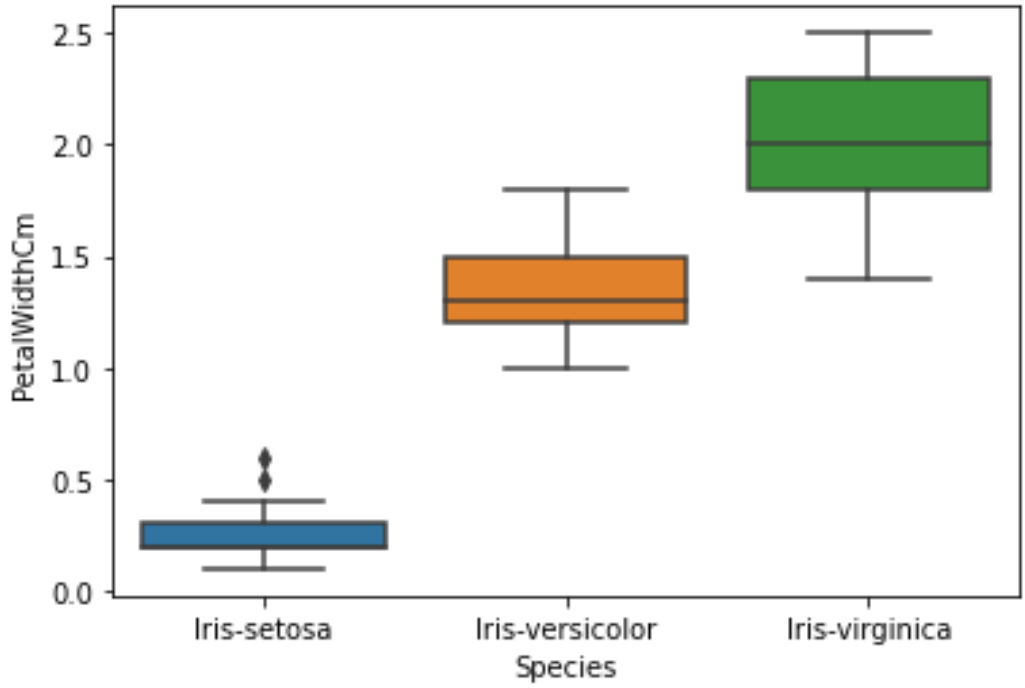

```
#Box plot for SepalLengthCm
sns.boxplot(x='Species',y='SepalLengthCm',data=df)
plt.show()
```




```
#Box plot for SepalWidthCm
sns.boxplot(x='Species',y='SepalWidthCm',data=df)
plt.show()
```




```
#Box plot for PetalWidthCm
sns.boxplot(x='Species',y='PetalWidthCm',data=df)
plt.show()
```

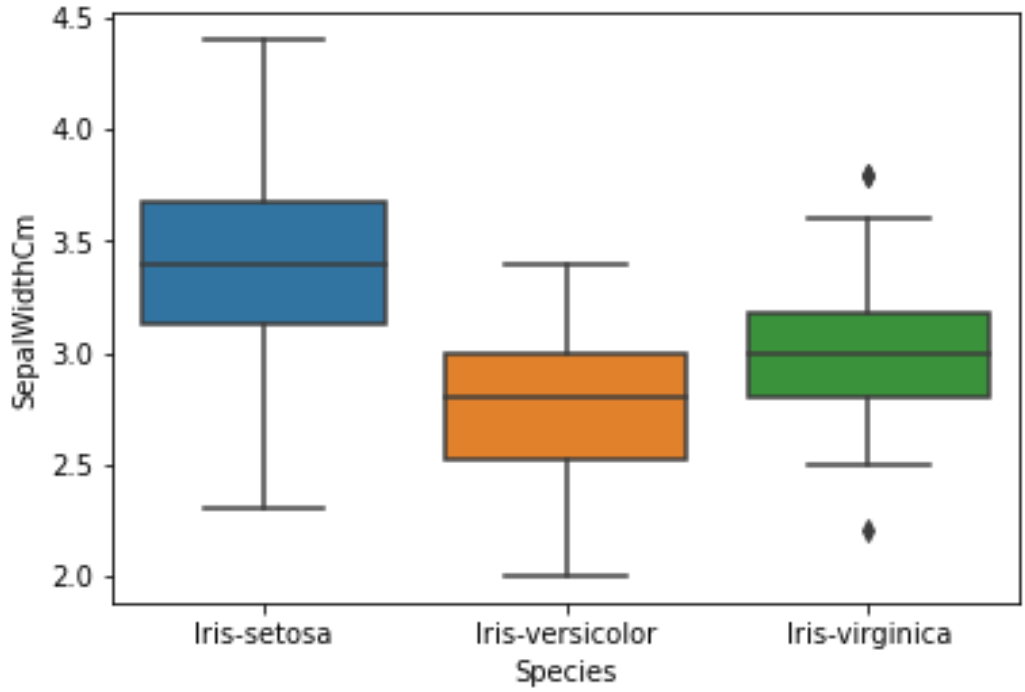


Observation

1. Outliers are observerd in petal Length for Setosa and Versicolor feature
2. Outliers are observerd in Sepal Length and Sepal Width feature for Verginica
3. Outliers are observed in Petal Width for Setosa

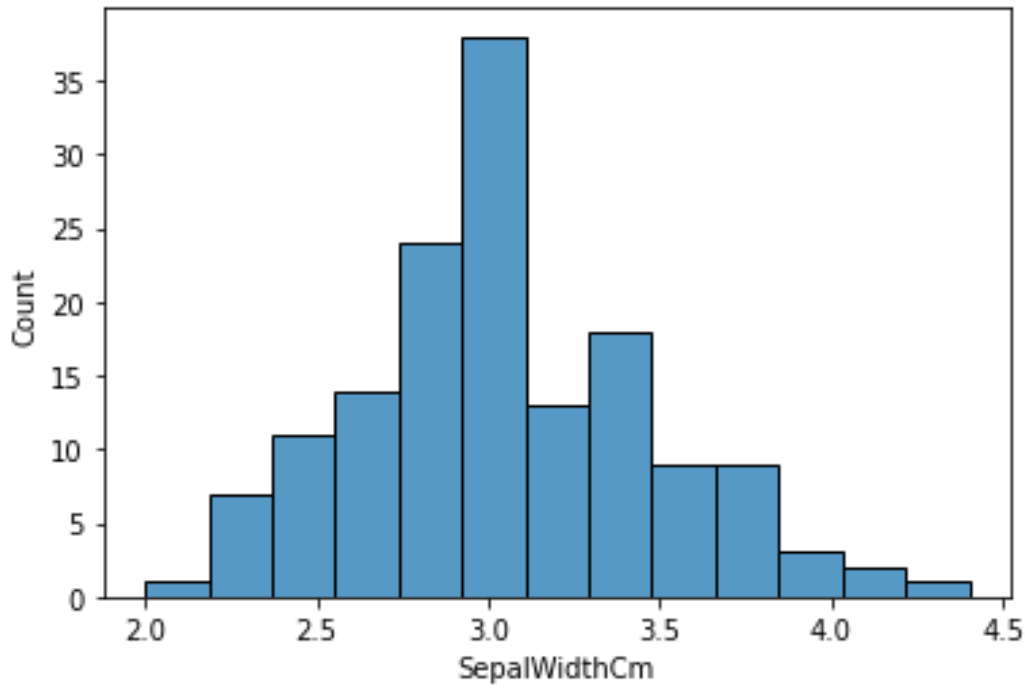
Generate box plot for a specific feature

```
# box plot for a special feature (sepal width)
sns.boxplot(x='Species',y='SepalWidthCm',data=df)
plt.show()
```



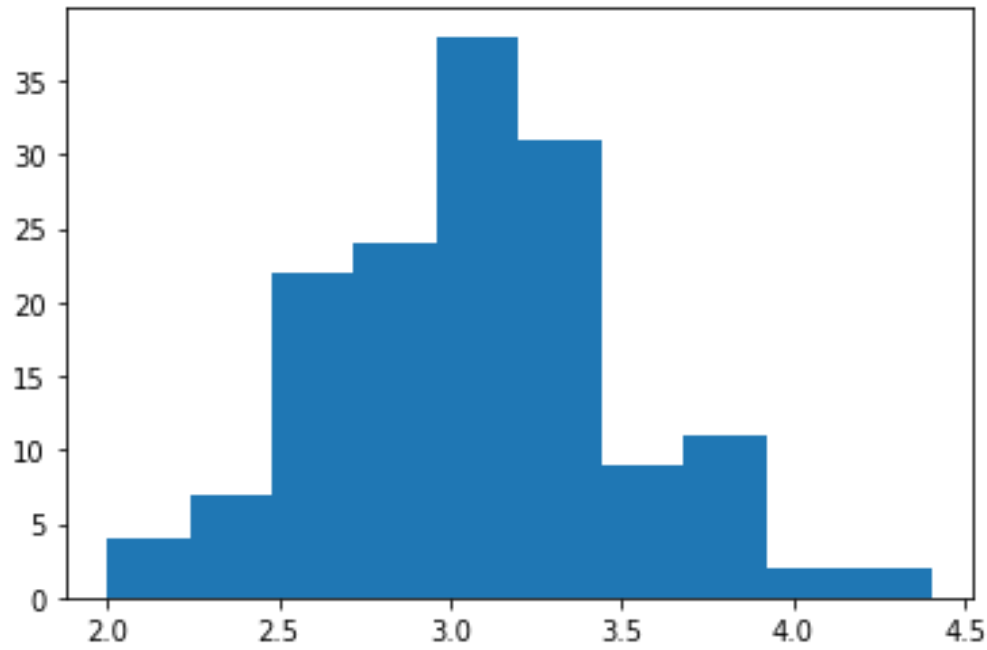
Generate histogram for a specific feature

```
# Generate histogram for a specific feature
sns.histplot(x='SepalWidthCm',data=df)
plt.show()
```




```
#!pip install -U seaborn
```

```
plt.hist(x='SepalWidthCm',data=df)
plt.show()
```



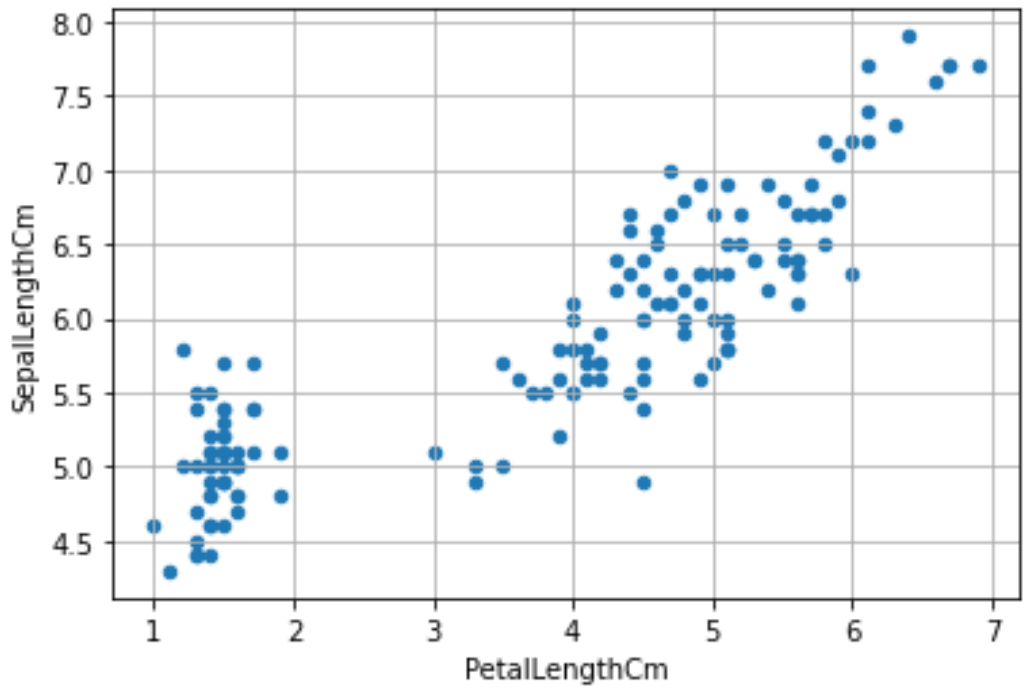

```
# !pip install seaborn --upgrade
```

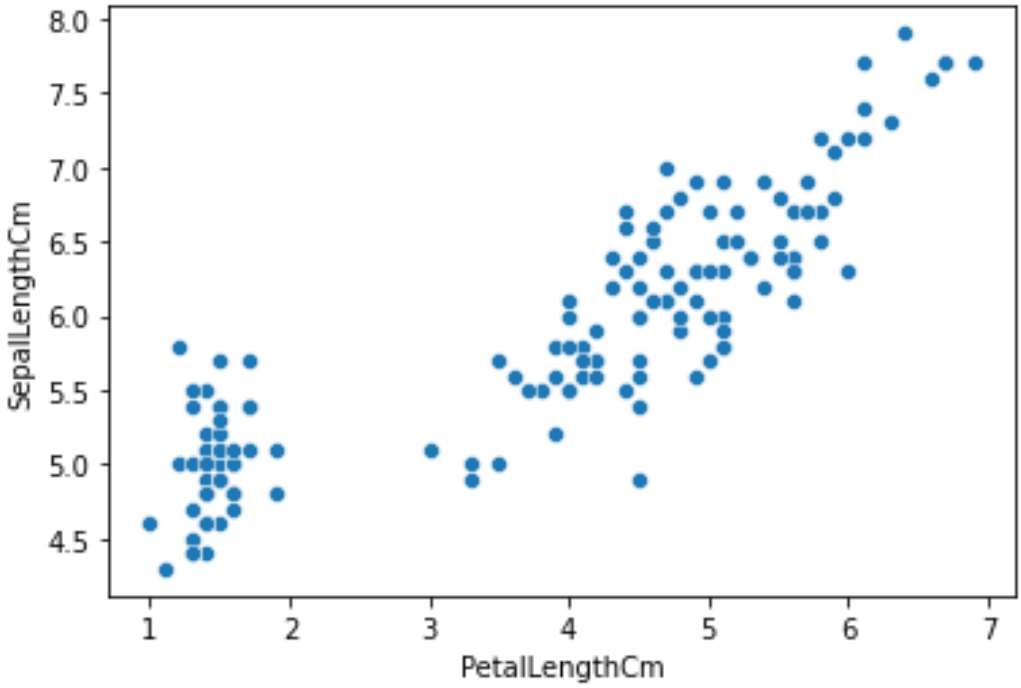
Observation

The given feature follows Continuous Probability Distribution

Generate Scatter plot of petal length vs. sepal length

```
#Scatter plot of petal length vs. sepal length
df.plot(kind="scatter",x='PetalLengthCm',y='SepalLengthCm')
#plt.show()
plt.grid()
```



```
sns.scatterplot(x='PetalLengthCm',y='SepalLengthCm',data=df)
# sns.set_style('darkgrid')
```

```
<matplotlib.axes._subplots.AxesSubplot at 0x2669b59fe80>
```